Example 20: What are the negations of the statements "There is an honest politician" and "All Americans eat cheeseburgers"?

Soln: Let $H(x)$ denote "x is honest." Then the statement "There is an honest politician" is represented by $\exists x\, H(x)$, where the domain consists of all politicians. The negation of this statement is $\neg \exists x\, H(x)$.
By De Morgan's law of quantifiers, $\boxed{\neg \exists x\, H(x) \equiv \forall x \neg H(x)}$.

In English, $\forall x \neg H(x)$ can be translated as, For all politicians, politician x is not honest, i.e. No politician is honest.

Let $C(x)$ denote "x eats cheeseburgers". Then the statement "All Americans eat cheeseburgers" is represented by $\forall x\, C(x)$, where the domain consists of All Americans. The negation of this statement is $\neg \forall x\, C(x)$.
By De Morgan's laws of quantifiers, $\boxed{\neg \forall x\, C(x) \equiv \exists x \neg C(x)}$.

In English, $\exists x \neg C(x)$ can be translated as "There exists an American who does not eat cheeseburgers."

Example 22: Show that $\neg \forall x (P(x) \rightarrow \neg Q(x))$ and $\exists x (P(x) \land \neg Q(x))$ are logically equivalent.

Soln: $\neg \forall x (P(x) \rightarrow \neg Q(x)) \equiv \exists x \neg (P(x) \rightarrow \neg Q(x))$ (By De Morgan's laws for quantifiers)

$\equiv \exists x \neg (\neg P(x) \lor Q(x))$ (By conjunction-disjunction equivalence)

$\equiv \exists x (\neg(\neg P(x)) \land \neg Q(x))$ (2nd De Morgan's law)

$\equiv \exists x (P(x) \land \neg Q(x))$ (Double-negation law)

1.4.10 Translating from English to Logical Expression

- There is no cookbook approach that can be followed step by step.

Example 23: Express the statement "Every student in this class has studied calculus" using predicates and quantifiers.